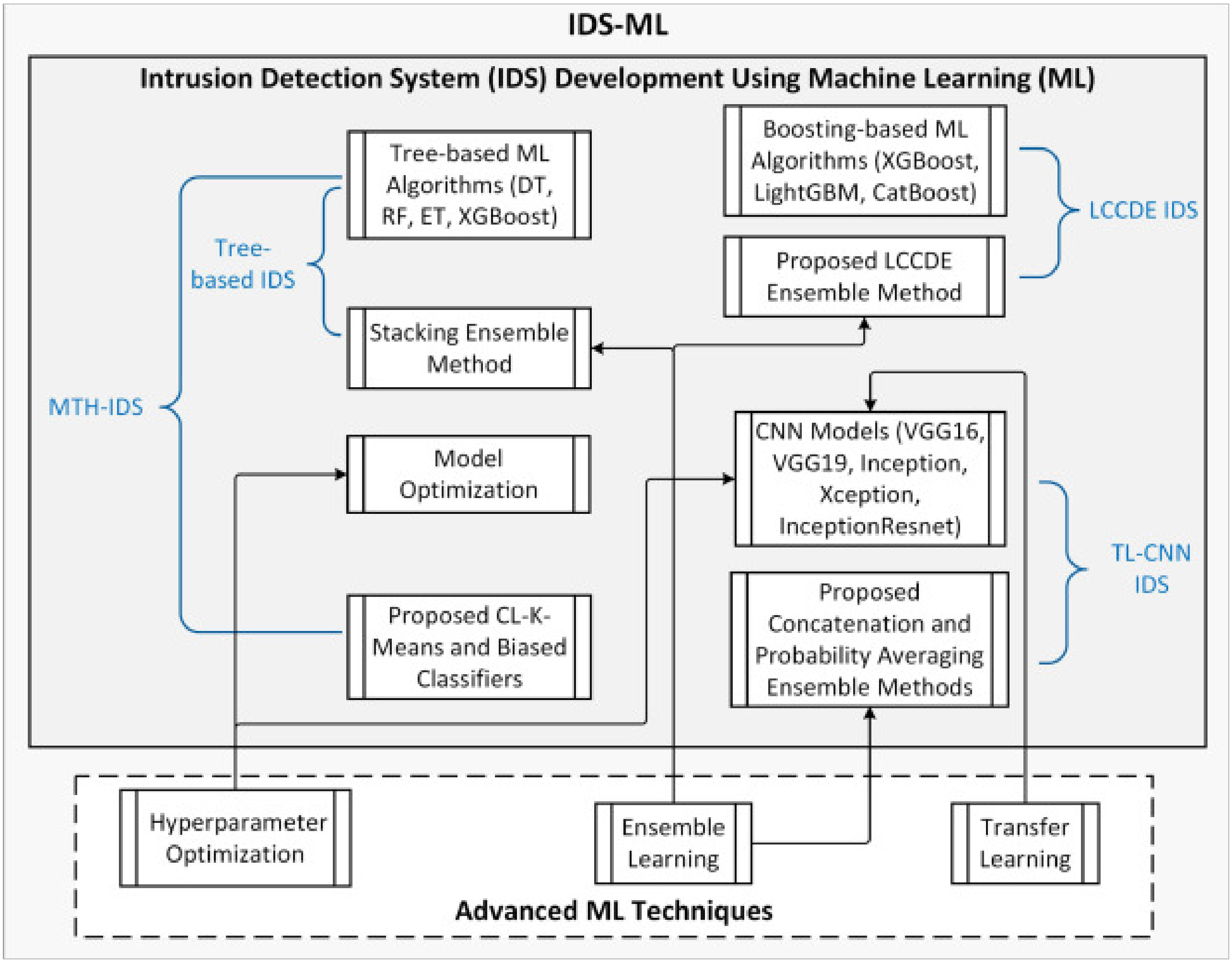

Figure 3: Machine learning-based IDS framework (IDS-ML)

Use case and deployment of an AI-driven IDS in Kubernetes

Deployment of an AI-driven IDS in Kubernetes involves the integration of eBPF-based monitoring with Snort or Suricata and machine learning models to identify anomalies.

Here are the deployment steps.

Set up eBPF monitoring configuration: Attach eBPF programs to network sockets to filter packets in real time.

Integrate Snort/Suricata: Grab network packets and pass them through predefined rule sets for analysis.

Feed data to AI models: Utilise machine learning to identify anomalies and enhance IDS accuracy.

Deploy in Kubernetes:

- Implement a Kubernetes DaemonSet to execute IDS agents on every node.
- Utilise Kubernetes Network Policies to apply security policies.
- Harness Prometheus and Grafana for security monitoring in real-time.

The IDS-ML model

The IDS-ML model uses various machine learning techniques for intrusion detection, including tree models (DT, RF, ET, XGBoost), CNN models (VGG16, VGG19, Inception, Xception, InceptionResnet), and ensemble techniques such as stacking and probability averaging. It also has optimisation techniques such as hyperparameter tuning and clustering-based classifiers for improved detection efficiency.

Here's the code for using YAML to deploy Snort in Kubernetes:

```
apiVersion: apps/v1
kind: DaemonSet
metadata:
    name: snort-ids
spec:
  selector:
    matchLabels:
      app: snort
  template:
    metadata:
      labels:
        app: snort
  spec:
    containers:
      - name: snort
        image: snort/snort:latest
```